

Lab 1 - Use-Case Flow Specification

UC-04: Prioritize ER Bed Allocation

Primary Actor	Triage Nurse
Supporting Actor	Hospital Administrator
Preconditions	The Triage Nurse is authorized to use the dashboard; patient intake information and valid vitals are available; current bed-status information is accessible.
Postconditions	The patient is assigned an appropriate position in the ER bed allocation waitlist (or an available bed is allocated), the decision is recorded, and the dashboard reflects the current allocation state.

Main Success Scenario

1. Triage Nurse selects the bed allocation function for a patient awaiting ER placement.
2. The system retrieves the patient intake vitals and current bed availability information.
3. The system computes the patient's triage severity score from the intake vitals.
4. The system compares the patient's severity with the severity of patients already on the ER waitlist.
5. The system dynamically orders the waitlist so higher-acuity patients are ahead of lower-acuity patients.
6. If a suitable ER bed is available, the system associates the patient with that available bed; otherwise the patient remains in the ordered waitlist.
7. The system records the allocation/waitlist decision and the relevant triage score.
8. The updated allocation and bed status are propagated to connected ER dashboard clients.
9. The use case ends successfully.

Alternate Flow - No Suitable ER Bed Available

- 6a1.** At Step 6, no suitable ER bed is available for the patient.
- 6a2.** The system keeps the patient in the dynamically ordered ER waitlist according to triage severity.
- 6a3.** The dashboard displays the patient's waitlist status for the authorized user.
- 6a4.** If the patient requires ICU transfer handling based on the triage priority, the patient is flagged for the ICU transfer workflow; the ambulance transfer workflow can then be initiated when inter-facility transfer is required.
- 6a5.** The use case returns to the main flow at Step 7.